

Devi Ahilya University
Institute of Engineering & Technology, Indore
Class Test: I

Sub Code: 4ETRC1
Class: II Year

Sub Name: Measurement System and Sensors
Branch: E & TC (A & B Sec)

Marks: 20

Duration: 1.10Hr.

Note: Attempt all questions. All questions carry equal marks.

- Q.1 Explain the terms precision, sensitivity, reproducibility, resolution and fidelity as used for indicating instruments.
- Q.2 How strain gauge is signal conditioning using (i) Ballast circuit and (ii) two strain gauges?
- Q.3 A Wheatstone bridge circuit uses four strain gauge of equal magnitude. These gauges have been bonded to fixed end of mild steel cantilever having dimension as 1m long X 100 mm wide X 30 mm deep. The nominal resistance of gauge is 200 Ω , $F=4$, supply voltage is 12V & measuring instrument has an infinitely high internal resistance. Calculate change in output voltage when 2Kg mass is applied at the free end of the cantilever and $E=400 \times 10^9 \text{ N/m}^2$.
- Q.4 A bimetallic strip consists of nickel chrome-iron alloy and invar bonded together at 50°C. The strip is 100mm long and each has thickness of 2mm. Calculate i) radius ii) angular displacement and iii) vertical displacement if strip is subjected to 400°C. Given that $\alpha_{\text{alloy}} = 12.5 \times 10^{-6} / ^\circ\text{C}$ and $\alpha_{\text{invar}} = 1.7 \times 10^{-6} / ^\circ\text{C}$.

$$F = \frac{\sigma}{E}$$

$$\epsilon = \frac{\sigma}{E}$$

$$\sigma = M$$

$$5M = m \times l$$

$$5y = \frac{x}{2}$$

$$5b = \frac{x}{12}$$

Devi Ahilya University
Institute of Engineering & Technology, Indore
Class Test: III

Sub Code: 4ETRC1
Class: II Year

Sub Name: Measurement System and Sensors
Branch: E & TC (A & B Sec)

Marks: 20

Duration: 1.10Hr.

Note: Attempt all questions. All questions carry equal marks.

- Q.1 Explain the Ultrasonic gauge for flow and level measurement.
- Q.2 How level is measured using (i) Gamma ray method and (ii) Resistive method?
- Q.3 How discharge is measured with Rectangular and Triangular Weir?
- Q.4 A venturimeter is used for measuring the flow of cooling water with the differential pressure between inlet & throat is 17.5 cm when measured with U-tube manometer. Make calculation for the diameter of throat if the discharge coefficient is 0.97, $\rho = 1000 \text{ Kg/m}^3$, specific gravity of mercury = 13.6. and inlet diameter is 15cm.

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Devi Ahilya University
Institute of Engineering & Technology, Indore
Class Test: II

Sub Code: 4ETRC1
Class: II Year

Sub Name: Measurement System and Sensors
Branch: E & TC (A & B Sec)

Marks: 20

Duration: 1.10Hr.

Note: Attempt all questions. All questions carry equal marks.

- Q.1 Explain the working of Bourdon gauge along with errors and their rectification.
- Q.2 Discuss the various section of venturimeter along its advantages & disadvantages. Also write the formula for discharge in case of venturimeter.
- Q.3 A force of 8N is impressed upon a piezoelectric crystal having dimensions of 6mm X 6mm X 1.5mm thick. For crystal materials, charge sensitivity is 140pC/N, permittivity is $12 \times 10^{-9} \text{ N/m}$ and modulus of elasticity is $11.5 \times 10^6 \text{ N/m}^2$. Calculate i) voltage generated ii) capacitance and iii) strain induced in the crystal.
- Q.4 A manometer is used to transfer oil of relative density 0.4 as the manometric liquid, but the scale is to be graduated in mm of water. If one leg is 1mm bore tube and other is 10mm diameter well, calculate the angle to the horizontal at which the tube & scale must be inclined to give 2.5mm scale deflection for a pressure difference equivalent of 0.5 mm head of water

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Subject: 4ETRG2 DIGITAL SIGNAL PROCESSING

Time: 70 Min

M. M. 20

Note: Attempt any four questions.

Q.1) What do you mean by Causality and Stability of system. Also explain [5]
with example how to determine in Z transform, if system transfer function is given.

Q.2) State and prove time and frequency shifting property in DTFT [5]

Q.3) Using the definition determine the DTFT of the following sequences. [5]
If it does not exist say why:

(a) $X[n] = 2^{|n|}$ (b) $X[n] = 2^n u[-n]$

Q.4) Compute the DFT of the following sequences [5]

(a) $[1, 0, -1, 0]$ (b) $[j, 0, j, 1]$

If (a) $x[n]$ is causal, also verify result using Residue method

Q.5) Find the stability and frequency response of DT-LTI system [5]
represented by the impulse response

(a) $h[n] = (1/2)^n u[n]$ (b) $h[n] = (1/8)^n u[n]$

II B.E. (Electronics & Telecommunication)

Class Test II March 24

Subject: 4ETRG2 DIGITAL SIGNAL PROCESSING

Time: 70 Min

M. M. 20

Note: Attempt any four questions.

Q.1) Determine and sketch Direct Form I and Direct Form II realizations [5]
For following pulse transfer function.

$$H(z) = (3 + 0.5z^{-1} + 0.7z^{-2} + 2z^{-3}) / (2 - 3z^{-1} - 4z^{-2} - 0.8z^{-3})$$

Q.2) (a) State and prove time reversal property in Z transform. [3]

(b) Determine Z transform of $(1/3)^n n^2 u[n]$ [2]

Q.3) Using Z transform, determine convolution of two sequences [5]

$$X_1[n] = [2, 1, 0, -1, 3] \text{ and } X_2[n] = [1, -3, 2]$$

Q.4) (a) Determine the discrete signal $x[n]$, whose z transform is given by [3]

$$X[z] = \log(1 + az^{-1}), \quad |z| > |a|$$

(b) State Initial and Final value theorem in z transform, [2]
calculate initial and final value corresponding $x[n]$,

$$\text{whose Z transform as } X[z] = 2 + 3z^{-1} + 4z^{-2}$$

Q.5) Using Long Division method, determine the Inverse Z transform of [5]

$$X[z] = \frac{1 + 2z^{-1}}{1 - 2z^{-1} + z^{-2}}$$

If (a) $x[n]$ is causal, also verify result using Residue method

II B.E. (Electronics & Telecommunication)

Class Test I February 24

Subject: 4ETRG2 DIGITAL SIGNAL PROCESSING

Time: 70 Min

M. M. 20

Note: Attempt any four questions.

Q.1 Express following signal in terms of step functions as well as sketch it. [5]
 $x(t) = \pi[(t-3)/6] + \pi[(t-4)/2]$

Q.2 Given the following signals: [5]

(a) $\cos(40\pi t - \pi/3) + \sin(50\pi t + \pi/6)$ (b) $\cos(30t) + \sin(60\pi t)$

(c) $u(t) - u(t-2)$ (d) $e^{-t}u(t)$

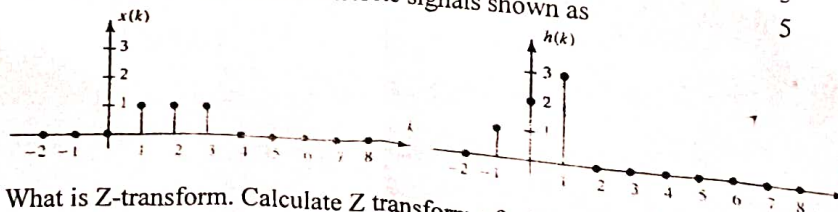
(e) $\sin^2(8\pi t - \pi/6)$

Which are periodic? determine their period.

Q.3 Find and the convolution of $x(t)$ and $h(t)$. [5]

$$x(t) = 2 * e^{(-10t)}u(t) \text{ and } h(t) = u(t-2)$$

Q.4 Find out convolution of two discrete signals shown as [5]
[5]



Q.5) What is Z-transform. Calculate Z transform of $x[n] = 5^n u[n]$, [5]
where $u[n]$ discrete step signal.

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II B.E. (E&TC – A&B)
Class Test II March 2024
Subject: 4ETRC2, Subject Name: EMF&TL

Duration: 70 min

Max Marks: 20

Note: Attempt any four questions.

- Q1** Transform each of the following vectors to spherical coordinates at the point specified 5
 (a) $5 \mathbf{a}_x$ at B $(3, 35^\circ, 120^\circ)$ (b) $5 \mathbf{a}_x$ at A $(3, 4, -2)$ (c) $7 \mathbf{a}_x - 4 \mathbf{a}_y - 7 \mathbf{a}_z$ at P $(-4, -5, 6)$
- Q2** If $\vec{A} = xz^3 \mathbf{a}_x - 2x^2yz \mathbf{a}_y + 2yz^4 \mathbf{a}_z$ find $\text{div } \vec{A}$ and $\text{curl } \vec{A}$ at the point $(2, -1, 3)$ 5
- Q3** A uniform line charge, infinite in extent with $\rho_l = 20 \text{ n C/m}$ lies along the Z axis. Find the Electric field at $(6, 8, 3)$ 5
- Q4** Calculate Electric field at P $(2, 2, 2)$ caused by four identical 5 n C charges located at $P_1(2, 2, 0)$, $P_2(-2, 2, 0)$, $P_3(-2, -2, 0)$ and $P_4(2, -2, 0)$ 5
- Q5** A Transmission line is lossless and 50 m long. It is terminated in a load of $Z_L = 50 + j40 \Omega$ at frequency of 20 MHz , the inductance and capacitance of the line are $L = 500 \text{ n H/m}$ $C = 60 \text{ p F/m}$. Find the input impedance at source and at the midpoint of the line. 5

II B.E. (E&TC – A&B)
Class Test III April 2024
Subject: 4ETRC2, Subject Name: EMF&TL

Duration: 70 min

Max Marks: 20

Note: Attempt any four questions.

- Q1** State and Prove Divergence theorem. 5
- Q2** Explain and derive the Biot-savart's law. 5
- Q3** Write the Maxwell's equation. Explain the physical significance of each equation. 5
- Q4** What is Smith chart? What are various applications of Smith chart in Transmission line? 5
- Q5** An UHF Transmission line operating at 1 GHz is connected to Z_L producing reflection coefficient $0.433 + j 0.25$. Calculate position of stub, length of stub and VSWR 5

II B.E. (E&TC – A&B)
Class Test I February 2024
Subject: 4ETRC2, Subject Name: EMF&TL

Duration: 70 min

Max Marks: 20

Note: Attempt any four questions.

- Q1** A Point charge $Q = 18 \text{ n C}$ is at the origin in free space. Find the electric field at point P $(3, 0, 3)$ also find the electric flux density at point P. 5
- Q2** Two point A $(3, 3, 2)$ and B $(4, -5, 3)$ are given in the Cartesian system obtain the vector from A to B, and a unit vector directed from A to B and also find angle between vector A to B 5
- Q3** Transform each of the following vector to cylindrical coordinates at the point specified 5
 (a) $7 \mathbf{a}_x$ at B $(\rho = 5, \phi = 135^\circ, z = -2)$ (b) $4 \mathbf{a}_x$ at A $(x = 3, y = 4, z = -2)$
 (c) $5 \mathbf{a}_x - 3 \mathbf{a}_y - 8 \mathbf{a}_z$ at P $(x = 3, y = 5, z = 7)$
- Q4** What is Transmission line? Derive the derivation of Voltage and Current equation of Transmission line. 5
- Q5** An open wire Telephone line has $R = 13 \text{ Ohm per meter}$, $L = 6.5 \times 10^{-7} \text{ Henry per meter}$ $C = 30 \times 10^{-12} \text{ Farad per meter}$ and $G = 0.26 \times 10^{-3} \text{ mho per meter}$ and operating frequency to 5 MHz find Propagation constant, characteristic impedance, attenuation constant and phase constant 5